optimal solution before trying all the possible (and reasonable) configurations.

Other times, getting to the right answer might not even be possible: an exponential algorithm becomes unfeasible for inputs larger than a few dozen elements, and we would not even be able to see the program output an answer on larger inputs.

Even a supercomputer that can perform ~10^{16} operations per second, to run an exponential algorithm that performs $O(2^n)$ steps, for an input size 100 will need $O(10^{14})$ seconds, which is, give or take, 3 million years! And if we were able to make our computers a million times faster (which, even when Moore's law was holding, would have taken 30 years), then we would increase our ability to solve the problem just a tiny bit: our computation would still require 3 years for 100 elements, 3 thousand years for 110, and again 3 million years for 120!

For easier problems, even if the time to wait would not be longer than human history, it could still be too long: think about live event, or forecasting – there won't be much use for tomorrow's weather forecast, if it's delivered in three days!

So, in all these cases, we might be willing to accept a sub-optimal answer, provided we can get it within a reasonable time – but that's not the whole story.

Another important consideration in favor of heuristics is that, despite not being able to guarantee the optimal solution on all inputs, some heuristics are able to return the optimal result, in a reasonable time, for a subset of the whole problem space, and in